

CSE321 - Introduction to Algorithm Design

HW03

Due date: 14 December 2015 Monday 23:59:59

Warning: You must write all of your algorithms in Python programming language. Otherwise, your work will not be graded.

Submission Policy: You can hand-in your solution to M. Rasih ÇELENLİOĞLU. Alternatively, you can compress your solutions and submit from Moodle. Name of the compressed file must be exactly *sur-name_name1_name2_studentID.file_extension* where all of the letters must be lowercase and there should be no Turkish characters.

(e.g., Mahmud Rasih ÇELENLİOĞLU – > celenlioglu_mahmud_rasih_123456789.zip)

- 1) Design a divide and conquer algorithm that finds the depth of a given binary tree. Analyze the complexity of your algorithm.
- 2) You have an unordered array A with size n . If there is such an element $A[i]$ which satisfies both of the conditions $A[i - 1] \geq A[i]$ and $A[i] \leq A[i + 1]$, we call $A[i]$ as valley. Design a divide and conquer algorithm that finds all the valleys in A . Analyze the complexity of your algorithm.
- 3) A number is *dominant* if it exists in more than half of the entries of an array. You are given an array $A[0..n]$ and your task is to design an efficient algorithm that finds the dominant number, if exists. Analyze the complexity of your algorithm.
- 4) Design both a brute force and a decrease and conquer algorithms that find the largest sum of consecutive numbers in an array. For example, let $A = \{-2, 11, -4, 13, -5, 2\}$. In this case, sub-array $A[2..4]$ is the largest ($A[2] + A[3] + A[4] = 20$). Analyze the complexities of your algorithms.